



Vendor Document Review Sheet (DECAL)

PROJECT TITLE:

1000 KTA POLYVINYL CHLORIDE (PVC)

VENDOR NAME:

KSB Limited

DOCUMENT TITLE:

Pump Datasheet for Recycled water pump (10kg/cm2g)

VENDOR DOCUMENT NO:

9975714953-100-PDS

VENDOR REVISION NO:

03

VENDOR DOCUMENT REVIEW FEEDBACK

Permission to proceed does not constitute acceptance or approval of Design Detail, Calculations, Analysis, Test Methods or Materials developed OR selected by Vendor, and does NOT relieve Vendor from FULL compliance with Contractual Obligations.



1 Approved.



2 Approved with comments. Work may proceed.



3 Revise and Resubmit. Work Shall not proceed.



4 Retained for Information.



5 Cancelled / Superseded



F Final Certified

By: _____

Date: _____

Discipline:

Mechanical

MOC Number:

NA

JI Number

N23P01

Equipment No.

MP-PP031-P600173A/B
MP-PP031-P700173A/B

MRQ Number

MT-N23P01-D11NPP-MPP001-TPD-Q-04

Program No

NP3701

MRP Number Excluding Discipline & Type

N23P01-D11NPP-MPP001-TPD-P-04

Unit No. (Plant WBS)

Cat. No

Seq. No.

Rev No

- 0 3 1 -

D 0 1 -

0 0 8 -

0 3



SI UNITS

JOB NO.	1	N23P01	ITEM NO.	MP-PP031-P600173A/B, P700173A/B	
PURCHASE ORDER NO.	1	C4C-6770323 / SQ202404046			
REQUISITION NO.	1	MT-N23P01-D11NPP-MPP001-TPD-P-04	DATE	13.03.2025	
REVISION NO.		03	BY		
PAGE	2	OF	4	2	

1	Applicable To:	☐ Proposal	☒ Purchase	☐ As Built	☐ ASME B73.1, B73.2	☐ API 610	☒ Other	ISO -5199 and MR-002
2	Manufacturer	KSB	Model	CPK 80-315	Type	OH1	No. Stages	1
3	No. Pumps Required	4		2	No. Motor Driven	4	2	No. Turbine Driven
4	Motor Data Sheet No.		Pump Item No.		MP-PP031-P600173A/B, P700173A/B		Pump Item No.	
5	Turbine Data Sheet No.		Motor Item No.	1	MP-PP031-M600173A/B, M700173A/B		Turbine Item No.	
6	Diesel Data Sheet No.		Motor Provided By		Pump Vendor		Turbine Provided By	
7	Gearbox Data Sheet No.		Motor Mounted By		Pump Vendor		Turbine Mounted By	

9	LIQUID		OPERATING CONDITIONS (to be read in conjunction with PDS)				SITE CONDITIONS				
10	Name	RECYCLED WATER	Capacity		m ³ /h	Temp (°C)	Max.	43	Min.	9.4	
11	Pumping Temperature	65 (°C)	Normal	100	Rated	120	Rel. Hum. (%)	Max.	96	Min.	15
12	Normal	Max	Min	Disch. Press (kg/cm ² g) (1)(2)		11.40		Altitude (m)			
13	Specific Gravity @:	°C	0.981	Suct. Pres. (kg/cm ² g) (1)(2)				☐ Indoor ☐ Heated ☒ Roof			
14	Vapour press (kg/cm ² a)@	°C	0.25	Max.	0.68	Rated	-0.10	☐ Outdoor ☐ Unheated ☐ Sun			
15	Viscosity (cP) @	°C	0.4	Diff. Pres. (kg/cm ²) (1)(2)		11.5		Area Classification			
16	Corrosion / Erosion Caused by	-	Diff. Head(m) (1)(2)		117		Other:				
17			NPSH Available (m) (1)(2)		7		Remarks:				
18	Remarks		Hyd. Power (kW)		37.52		①				
19			☐ Parallel ☐ Series Operation								
20			☒ Continuous ☐ Intermittent Service								

21	PERFORMANCE (VTA)										2
22	Proposal Curve No.	K 2721.452/345/6	1	Min. Continuous Flow	(m ³ /h)	Rated Power (kW)	56				
23	Efficiency (%)	67.02%		Thermal 25	Stable 30	Max. Power @ Rated Imp. (kW)	68				
24	NPSH Required (m Water)	2.8		Suction Specific Speed	12794	rpm, m, m ³ /h	Driver kW	75			
25	Max. Head @ Rated Imp. (m)	128.5		Impeller Dia. (mm)			Driver Speed (rpm)	2972			
26	Max. Head @ Max. Imp. (m)	148		Rated 300	Max 320	Min 268	Pump Speed (rpm)	2972			

CONSTRUCTION (VTA)

29	MAIN CONNECTIONS								
30		SIZE	RATING	FACING	POSITION	MECHANICAL SEAL:			
31	Suction	5"	150 #	RF	HORIZONTAL	API Class Code	API 682	Edit	4th
32	Discharge	3"	150 #	RF	VERTICAL	Manufacturer	EagleBurgmann		
33	Bal. Drum	-	-	-	-	Model	APITEX-045-SG00-IN004A2		
34						Manufacturer Code	Q101VMG		

35	OTHER CONNECTIONS						Cartridge Mount	Single
36		No.	SIZE	RATING	TYPE	POSITION	Hook Sleeve or Non Cartridge	NA
37	Drain	1	1/2"	150 #	RF	HORIZONTAL	Pumping Ring	NA
38	Vent	-	-	-	-	-	Aux Sealing Device	NA
39	Press. Gauge	-	-	-	-	-	PACKING:	
40	Temp. Gauge	-	-	-	-	-	Packing Mfg. / Type	N/A

41	CASING MOUNTING					Size / No. Rings	N/A	
42	☒	Foot	☐	Centreline	☐	Near Centreline	☐	Inline
43	☐	Sump	☐	Vertical	☐	Vertical Barrel	☐	Bracket

GLAND PLATE TAPS REQUIRED:

☒	Quench	☒	Flush	☒	Drain	☐	Vent
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44	CASING SPLIT	☐ Cooling	☐ Heating	☐ Barrier
45	☒ Radial ☐ Axial	BEARINGS (SIZE / TYPE)		

46	CASING TYPE				Radial	Deep groove ball bearing 06409 C3
47	☒	Single Volute	☐	Double Volute	Thrust	Deep groove ball bearing 06409 C3
48	☐	Staggered	☐	Vertical Double	LUBRICATION TYPE:	N/A
			☐	Barrel		

49	IMPELLER DESIGN				☐	Grease	☐	Flood	☐	Pure Oil Mist
50	☐	Impeller Style:	☐	Open ☒ Closed	☒	Ring Oil	☐	Flinger	☐	Purge Oil Mist
51	☐	Impeller Mount:	☐	Btwn brgs. ☒ Over hung	☐	Pressure	☐	API 614	☐	API 610

52	Shaft Rotation (Viewed From Coupling End)		COUPLING	
53	☒	CW	☐	CCW
			Manufacturer	RATHI

54	SEALING SYSTEM:			Type	FLEXIBLE METALLIC SPACER SWQ-276(140) STD CPLG	2
55	☐	Mag-Drive	☒ Mechanical Seal	☐	Packing	
				☒ Coupling Guard	☒ Non-Sparking	☐ Other N/A

56	Spacer length shall be adequate to allow for seal removal
57	VALVES

58	☒	Casing Drain	☐	Purchaser	☒	Manufacturer	☐	Make	<u>KSB</u>	☐	Model	<u>NA</u>
59	☐	Casing Vent	☐	Purchaser	☐	Manufacturer	☐	Make		☐	Model	

4	60	DRIVER
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61	DRIVER TYPE	Motor	ORIENTATION	Horizontal
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62	VARIABLE SPEED REQUIRED	☐ Yes	☒ No	STARTING METHOD	Open Valve
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62	WARRANTY OF LBS REQUIRED	—	—	—	—	STARTING METHOD	Open Valve
63	MANUFACTURER	CGI				NOMINAL RPM	2972

64	NAMEPI ATE POWER	75 K _w	RATED LOAD RPM	2972
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64	NAME PLATE POWER	75 KW	RATED LOAD KVA	277.2	SEE DRIVER DATA SHEET
65	FRAME OR MODEL	280S			

66	FORMER MODEL	2005	SEE DRIVER DATA SHEET	
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[illegible]

Recycled Water Pump (10 kg/cm ² g)	
DATA SHEET No.	REV

		DATASHEET NO.	REV
		N73P01-D11NPP-MPP001-TPD-P-04	

CENTRIFUGAL PUMP DATA SHEET

CENTRIFUGAL PUMP DATA SHEET

(2)

051-D01-000

SHEET	2	OF	4
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JOB NO. 1 N23P01 ITEM NO. MP-PP031-P600173A/B, P700173A/B

PURCHASE ORDER NO. 1 4C-6770323 / SQ20240406

REQUISITION NO. MT-N23P01-D11NPP-MPP001-TPD-P-04 DATE 13.03.2025

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CENTRIFUGAL PUMP DATA SHEET

SI UNITS

MATERIALS (TO BE COMPLETED BY BUYER & SELLER)

Table H-1 Class	Imp. Wear Rings	<u>A743 Gr. CF3</u>	<u>2</u>	Coupling Hubs
Barrel/Casing	Case Wear Rings	<u>A743 Gr. CF3 (Coat)</u>	<u>3</u>	Coupling Spacer
Case	Shaft	<u>A276 TP 410 COND H</u>		Coupling Diaphragms
Impeller	Sleeve	<u>316L</u>		Coupling Collars
Remarks	Gland	<u>316L</u>		Bearings

BASEPLATE

Baseplate	☒ Yes ☐ No	☐ Baseplate Prepared with grout holes & vent holes
API Standard Baseplate No.		☐ Rain Gutter ☒ Drip Pan ☒ Baseplate Drain Connection
Type: ☒ Fabricated Steel ☐ Cast Iron		☐ Removeable Jack Screws for driver Horizontal Allignment
Other (Plastic)		☐ Four Point Lifting Lugs

*Remarks: Baseplate required under pump and motor

SEAL AND AUXILIARY PIPING

Primary Flush Plan	<u>Plan 11</u>	Secondary Flush Plan	<u>61M</u>	Cooling Water Piping Plan	
☐ Tubing	☐ Carbon Steel	☐ Tubing	☐ Carbon Steel	☐ Galvanised Pipe	
☒ Pipe	☒ Stainless Steel	☒ Pipe	☒ Stainless Steel	☐ Copper Tubing	☐ SS Tubing
Piping Assembly:		Piping Assembly:		☐ Sight Flow Indicators	
☐ Seal Welded	☒ Socket Welded	☐ Seal Welded	☒ Socket Welded	Cooling Water:	
☐ Threaded	☒ Flanged ☐ Union	☐ Threaded	☒ Flanged ☐ Union	Rated Flow(m³/h)	
☐ Type Tube Fitting:		☐ Type Tube Fitting:		Rated Pressure (kg/cm²a)	
External Flush Fluid	<u>NA</u>	External Flush Fluid	<u>NA</u>		
Specific Gravity		Specific Gravity		Notes:	
Specific Heat		Specific Heat			
Flow Rate Max./Min		Flow Rate Max./Min			
Pressure Max/Min		Pressure Max/Min (bar g)			
Temperature Max/Min		Temperature Max/Min (°C)			

INSTRUMENTATION

TEMPERATURE, PRESSURE AND LEVEL SWITCHES		VIBRATION AND BEARING TEMPERATURE DETECTION (RTD)			
☐ Temperature Gauge	NA	☐ Noncontacting Probes (API 670)		☐ Radial RTD	
☐ Thermowell	NA	☐ Accelerometer Probes		☐ Thrust RTD	
☐ Pressure Gauge	NA	MAG-DRIVE	☐ No	☐ Yes	
☐ Pressure Switch	NA	☐ Low Amp. Relay	☐ Power Monitoring System		
☐ Level Switch	NA	Containment Shell: ☐	"J" T/C ☐	"K" T/C ☐	RTD ☐
☐ Level Switch					

INSPECTION AND TESTING (shall be as per Inspection class 0 of MR002)

TEST	NON-WIT	WITNESS	ITEM	Mag particle	Dye Penetrant	Radio graphic	Ultrasonic
Perform.	☐	☒					
Hydrostatic	☐	☒	Nozzle	☐	☐	☐	☐
NPSH	☐	☐	Casting	☐	☒	☐	☐
Complete Unit Test	☐	☐					
Sound Level Test	☐	☒					
Pipe Load Test	☐	☐					

Max. Allowable Working Pressure (MAWP): 16.5 (kg/cm²g) 1

Hydrotest Pressure (kg/cm²g): 24.75 1

MASSES and DIMENSIONS

Mass of Pump	<u>125</u> kg	Mass of Pump	<u></u> kg	Dimensions:	
Mass of Baseplate	<u>325</u> kg	Mass of Baseplate	<u></u> kg	Baseplate for Motor-driven Pump	
Mass of Motor	<u>600</u> kg	Mass of Turbine	<u></u> kg	(L*W*H) <u>2200 mm X 720 mm X 150 mm</u> <u>1</u>	
Mass of Gear	<u>NA</u> kg	Mass of Gear	<u></u> kg	Baseplate for Turbine-driven Pump:	
Total Mass	<u>1050</u> kg	Total Mass	<u></u> kg	(L*W*H)	

PUMP ALLOWABLE NOZZLE LOADS (Force N, Moment Nm) AS APPROVED GAD

(N)	FX	FY	FZ	MX	MY	MZ
Suction Nozzle						
Discharge Nozzle						

REMARKS The nozzle loads and moments shall follow the RPMS-ENG-SPC-MR-002

NOTES

Recycled Water Pump (10 kg/cm2g)

CENTRIFUGAL PUMP DATA SHEET

N23P01-D11NPP-MPP001-TPD-P-04-031-D01-008 SHEET

JOB No N23P01

DATA SHEET No.

N23P01-D11NPP-MPP001-TPD-P-04-031-D01-008

REV

03

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REVISION

REVISION